

Societal Internship Summary Report on

COMMUNITY-BASED RAINWATER HARVESTING & DRIP IRRIGATION MODEL

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Duration: 15 Days

1. Introduction & Current Problem

The agricultural sector in Gujarat heavily relies on a consistent and predictable water supply to sustain the rural economy. However, regions like the Mehsana district fall under semi-arid zones, which are characterized by highly erratic rainfall and extremely limited surface water reservoirs. This geographical reality has forced the local farmers to rely almost exclusively on deep groundwater extraction. Over the years, the rapid and unregulated over-extraction of water using high-powered submersible pumps has led to a critical and alarming decline in the water table, officially classifying the region as "over-exploited." In Mathasur village, farmers are caught in a vicious financial cycle—as water levels drop further each year, they are forced to drill even deeper borewells (often exceeding 600 to 800 feet), which dramatically increases their electricity costs and pushes them into debt. The unpredictable monsoon rains frequently cause sudden surface floods that fail to naturally recharge these deep aquifers, leaving the farmers incredibly vulnerable during crucial crop sowing seasons. There is an urgent, undeniable need for grassroots-level, sustainable water management to secure the livelihood of these rural communities.



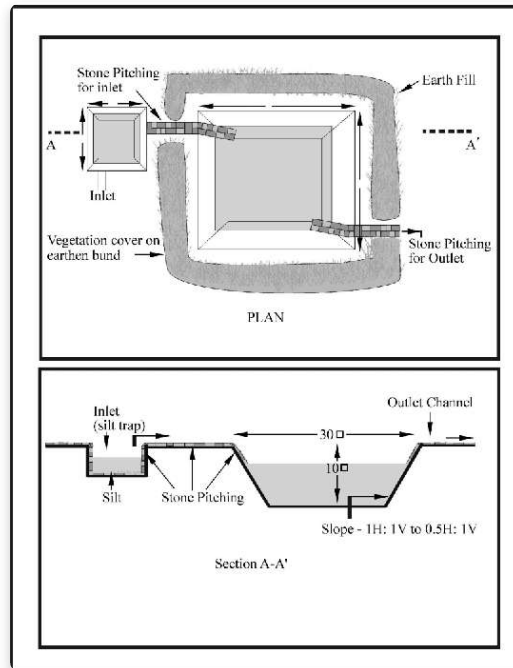
2. Core Objective of Internship

The core objective of this societal internship was to thoroughly investigate the ongoing water crisis in Mathasur village and to design a localized, community-driven solution that is both technically feasible and economically viable for small-scale farmers. The project aimed to bridge the widening gap between traditional, water-heavy farming methods and modern sustainable practices by proposing a carefully integrated model of Rainwater Harvesting (RWH) and Micro-Irrigation. By doing so, the project intended to significantly reduce groundwater dependency, halt the dangerous depletion of the water table, and lower the exorbitant electricity costs currently borne by the farming households. Furthermore, a highly significant objective of this fieldwork was to raise active awareness among the village stakeholders regarding the long-term economic benefits of proper resource management. This included educating and guiding them on how to navigate and leverage government subsidies, such as the PMKSY scheme, for adopting advanced drip irrigation technologies.

①	Q: What is the total area of your farmland?
	Answer: 4 Bigha (Approx. 1.6 Hectares)
②	Q: What is the current depth of your active borewell, and how much has it dropped in the last 5 years?
	Answer: • Current depth is around 650 feet. • It has dropped by almost 100 feet in the last 5 years. • Two old borewells have already dried up.
③	Q: What crops do you primarily grow in Kharif and Rabi seasons?
	Answer: • Kharif: Cotton (Kapas) and Castor (Erandi). • Rabi: Wheat and Mustard.
④	Q: Are you aware of the PMKSY micro-irrigation subsidy?
	Answer: Heard about it from the Sarpanch, but the paperwork seems very complicated and I am worried about the initial cost.
⑤	Q: What is your approximate monthly electricity bill for operating the submersible pump?
	Answer: During peak summer/dry season, it goes up to ₹3,500 - ₹4,000 per month just for pumping water.

3. Field Work Details & Execution

The execution of the project began with a comprehensive geographical and topographical physical survey of the Mathasur farmlands to accurately map the natural slopes and determine surface runoff directions. This essential step was immediately followed by intensive on-ground fieldwork involving structured interactions, detailed interviews, and informal group discussions with local farmers and the village Sarpanch. Critical data regarding existing borewell depths, seasonal crop water requirements (specifically for cotton, castor, and wheat), and average monthly electricity expenditures were meticulously documented. Based entirely on this primary data, a targeted hybrid solution was designed. The proposed design included Farm Ponds (Khet Talavadi) situated precisely at the lowest elevation points to naturally capture rainwater, lined with HDPE geomembranes to prevent rapid percolation. This was paired with deep recharge shafts containing natural filtration layers, and a localized drip irrigation layout plan to ensure that every single drop of pumped water is delivered directly to the crop roots, completely minimizing evaporative and surface loss.



Stakeholder Feedback Summary

Stakeholder Group	Primary Concerns & Feedback
Local Farmers	Declining water levels, high electricity bills, lack of capital for new technologies, and fear of complex subsidy paperwork.
Village Sarpanch	Need for communal water storage, willingness to allocate panchayat land for check-dams, request for technical guidance.
Agriculture Extension Workers	Emphasized the need for micro-irrigation training. Noted that farmers mistakenly believe drip irrigation is unsuitable for cash crops like cotton.

4. Conclusion & Future Impact

The proposed integration of scientific rainwater harvesting and localized drip irrigation serves as a highly effective, realistic blueprint for combating the severe water scarcity in Mehsana's agrarian communities. The financial feasibility analysis clearly demonstrated that while the initial capital investment for infrastructure might seem daunting to a local farmer, the return on investment is achieved rapidly through massive long-term savings in electricity and water pumping costs, especially when heavily supported by available government subsidies. The long-term environmental impact of this model is profound: actively and continuously recharging the aquifers will stabilize the local water table, improve general soil health by preventing topsoil erosion, and firmly secure crop yields against unpredictable monsoons. Ultimately, this community-based model not only addresses the immediate, dire crisis in Mathasur village but also successfully provides a scalable, sustainable framework that can be easily replicated across similar semi-arid regions in Gujarat, ensuring absolute water security for future generations.